Same-exposure ROI ratio diagnostic
Mathematics and implementation, version 1

Wenu-imaging

4 November 2026

Purpose and measurement stage

The diagnostic compares two spatial regions within each individual flat exposure: `near_readout` (numerator) and `parallel_far` (denominator). Their configured active-coordinate rectangles share the same x range and differ in y. It examines the observed readout-side bifurcation; it does not estimate a shutter coefficient or fit a detector response.

The current input is the retained raw-pair ASDF archive, bias-subtracted *without overscan correction*. This stage is propagated into every ratio product and displayed in HTML. The owner requires a subsequent consistent overscan subtraction for all bias and flat frames before interpretation of corrected linearity. That measurement pass is separate: no images are read, no previous measurements replaced, and no line fit recomputed here.

Identity and equations

For each detector, match samples by original input index, asset name, HDU, pair identifier and pair member. Require equal exposure time and detector. Time alone is not an identity: the two pair members remain separate. Duplicate identities or unequal ROI populations are rejected. With finite positive bias-subtracted ROI signals N_j, D_j (ADU), form

$$R_j = \frac{N_j}{D_j}. \tag{1}$$

Missing, nonfinite or nonpositive signals yield an unavailable ratio while preserving the input samples and reason. Let the inclusive reference bounds be $L = 15000$ and $U = 35000$ ADU, explicitly recorded with the product. Define

$$\mathcal{J} = \{j : R_j \text{ available},\ L \le N_j \le U,\ L \le D_j \le U\}, \tag{2}$$

$$R_{\text{ref}} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} R_j, \tag{3}$$

$$Q_j = R_j / R_{\text{ref}}, \qquad P_j = 100(Q_j - 1). \tag{4}$$

The reference is the unweighted mean of individual ratios, not a ratio of averaged signals. Save its value, count and exact member identities. If the reference is unavailable, keep raw ratios and leave Q_j, P_j unavailable. All exposures, including those outside the reference bounds, remain retained.

Interpretation and limits

In a linear detector with a stable illumination pattern, write $N_j = k_N F_j T_j$ and $D_j = k_D F_j T_j$, where T_j is effective exposure time and F_j is common illumination. Then $R_j = k_N/k_D$: both common brightness and common timing cancel. The normalization removes the constant spatial response ratio without applying a flat-field correction to the measurements.

Residual additive terms do not cancel:

$$R_j = \frac{k_N F_j T_j + B_{N,j}}{k_D F_j T_j + B_{D,j}}. \tag{5}$$

Consequently an overscan or bias mismatch can be particularly conspicuous at low signal. Changing illumination geometry, spatial response differences and detector nonlinearity can also produce a trend. The ratio is diagnostic, not a unique attribution to one mechanism. Common multiplicative cancellation requires the assumptions above. No spatial shutter model is introduced.

No uncertainty or statistical weight is fabricated. Normalized points share R_{ref} and therefore are correlated. This version records uncertainty as unavailable; quantitative inference needs an explicit error/covariance model.

Connection to software

`pycavi_roi_ratio.compute` implements matching, reference selection and the equations as a pure function. It retains deep copies of both original samples, units, availability, settings and membership. It uses finite checks and a stable summation for the reference.

`pycavi_roi_ratio_qc.publish` reads saved residual-ASDF original rows, freezes the source graph locally, and writes a new immutable QC product using the unchanged common factory, writer and reader. The operation is `diagnostic.same_exposure_roi_ratio`. It records source record digests, byte hashes, producer revision and processing stage. Roundtrip equality and source preservation are required. Broken links or identity mismatches fail.

`tools/compute-roi-ratios.py` consumes a completed source index and writes fresh per-detector products plus a completion index. Expected real coverage is 32 detectors, 20 matched exposures each: 640 ratio rows. `tools/view-pair-qc.py --ratios` validates and embeds those products; HTML only selects stored values. Its axes/cursor use the shared inspection module. No scientific division, normalization or line fitting occurs there.

Verification

Eight storage/computation tests cover analytic ratios and reference membership, common-factor cancellation, exact member identity, unavailable cases, malformed populations, relocation, frozen links, immutable sources and separate CLI completion. Viewer checks disable both numerical services while loading saved products. JavaScript checks exercise ratio modes, exact source inspection and shared axes/cursor hooks. Actual Mac qualification and real-data ratio publication are separate acceptance gates; fixture success is not scientific validation of the detector or a shutter model.